

5. Project Development Phase

Introduction

The development phase involved implementing the complete machine learning workflow, from data preprocessing to deployment of the trained model using Flask.

Development Steps

Step 1: Environment Setup

Installed required libraries:

- * NumPy
- * Pandas
- * Matplotlib
- * Seaborn
- * Scikit-learn
- * Flask
- * Pickle

Step 2: Dataset Collection

Downloaded the Human Development Index dataset and stored it in the dataset folder.

Step 3: Data Preprocessing

- * Loaded dataset using Pandas
- * Removed unnecessary columns
- * Handled missing values
- * Selected required features

Step 4: Data Visualization

Created graphs including:

- * Strip Plot

- * Correlation Heatmap

- * Distribution Analysis

Step 5: Model Development

Algorithm Used:

Linear Regression

Performed:

- * Train-Test Split

- * Model Training

- * Model Evaluation

Step 6: Model Saving

Saved the trained model as:

model.pkl

using Pickle.

Step 7: Flask Web Application

Created:

- * app.py

- * index.html

- * result.html

Implemented prediction functionality.

Project Structure

dataset/

models/

src/

templates/

static/

app.py

requirements.txt

README.md

Output

The Flask application accepts user input and predicts the Human Development Index accurately using the trained machine learning model.

Conclusion

The development phase successfully integrated machine learning and web technologies into a complete HDI prediction system.